

Course Code	21CSC305P	Course Name	MACHINE LEARNING	Course Category	C	PROFESSIONAL CORE										L	T	P	C	
																2	1	0	3	
Pre-requisite Courses	Nil		Co- requisite Courses	Nil		Progressive Courses	Nil													
Course Offering Department		School of Computing			Data Book / Codes / Standards			Nil												
Course Learning Rationale (CLR):		The purpose of learning this course is to:				Program Outcomes (PO)												Program Specific outcomes		
CLR-1:	Explore the fundamental mathematical concepts of machine learning algorithms					1	2	3	4	5	6	7	8	9	10	11	12			
CLR-2:	Apply linear machine learning model to perform regression and classification					Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CLR-3:	Utilize mixture models to group similar data items																			
CLR-4:	Develop machine learning models for time –series data prediction																			
CLR-5:	Design ensemble learning models using various machine learning algorithms																			
Course Outcomes (CO):		At the end of this course, learners will be able to:																		
CO-1:	Understand the basics of machine learning using probability theory					-	2	-	-	-	-	-	-	-	-	-	-	1	-	-
CO-2:	Implement machine learning models using supervised learning algorithms					-	2	-	2	-	-	-	-	-	-	-	-	-	-	2
CO-3:	Implement machine learning models using unsupervised learning algorithms					-	3	-	3	-	-	-	-	-	-	-	-	-	-	2
CO-4:	Implement machine learning models for sequential data analysis and prediction					-	3	-	3	-	-	-	-	-	-	-	-	-	-	3
CO-5:	Develop ensemble learning models for supervised and unsupervised learning					-	3		3	-	-	-	-	-	-	-	-	-	-	3
Unit-1 - Introduction9 Hour																				
machine learning what and why?, supervised and unsupervised learning, polynomial curve fitting, probability theory- discrete random variables, fundamental rules,Bayes rule, Independence and conditional independence, continuous random variables, Quantiles, Mean and variance, probability densities, Expectation and covariance.																				
Practice:																				
1. Devise a program to import, load and view dataset																				
2. Create a program to display the summary and statistics of the dataset																				
Unit-2 - Linear models for regression9 Hour																				
Maximum likelihood estimation – least squares, robust linear expression, ridge regression, Bayesian linear regression. Linear models for classification: Discriminant function – Probabilistic generative models, Probabilistic discriminative models, Laplacian approximation, Bayesian logistic regression, Kernels functions, using kernels in GLMs, Kernel trick, SVMs.																				
Practice:																				
1. Implement linear regression to perform prediction																				
2. Implement Bayesian logistic regression and SVM for classification																				
Unit-3 - Mixture models and EM9 Hour																				
K-means clustering, mixtures of Gaussians, An alternative view of EM, Factor analysis, PCA, choosing the number of latent dimensions. Clustering – measuring dissimilarity, evaluating the output of clustering methods, Hierarchical clustering.																				
Practice:																				
1. Implement K-means clustering, mixtures of Gaussians and Hierarchical clustering algorithm to categorize data.																				
2. Create a program to perform PCA																				

Unit-4 – Hidden Markov Models	9 Hour
Sequential data – Markov models, HMM – maximum likelihood for the HMM, The forward and Backward algorithm, the sum-product algorithm, scaling factors, Viterbi algorithm, linear dynamical systems.	
Practice:	
1. Implement HMM to predict the sequential data	
Unit-5 – Combining Models	9 Hour
Bayesian model averaging, Boosting, Adaptive basis function models, CART, generalized additive models, Ensemble learning.	
Practice:	
1. Implement CART learning algorithms to perform categorization	
2. Implement Ensemble learning models to perform classification	

Learning Resources	1. Pattern Recognition and Machine Learning, Christopher M Bishop, Springer, 2006. 2. Machine Learning- A probabilistic perspective, Kevin P.Murphy, The MIT Press, 2012.
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Learning Assessment									
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)						Final Examination (0% weightage)	
		Formative CLA-1 Average of unit test (20%)		Project Based Learning CLA-2 (60%)		Report and Viva Voce (20% weightage)			
		Theory	Practice	Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	-	15%	-	15%	-	-
Level 2	Understand	25%	-	-	20%	-	20%	-	-
Level 3	Apply	30%	-	-	25%	-	25%	-	-
Level 4	Analyze	30%	-	-	25%	-	25%	-	-
Level 5	Evaluate	-	-	-	10%	-	10%	-	-
Level 6	Create	-	-	-	5%	-	5%	-	-
	Total	100 %		100 %		100 %		-	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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